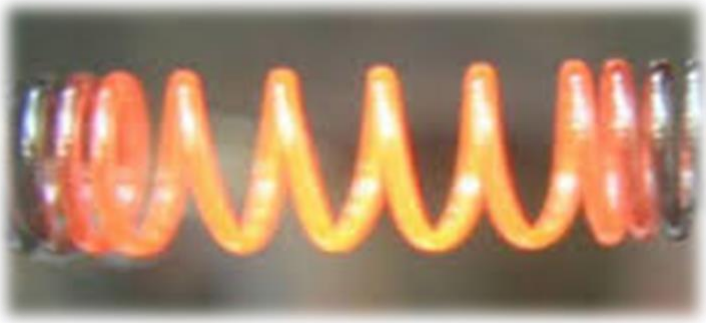
MIPS

Modular Intelligent Power Sources

Filament Module
Operations Manual Addendum
Rev 3.0, October 16, 2016



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WARNING

Only qualified electrical workers should operate and install this system.
Avoid spilling liquids onto/into the unit.
Do not expose to excessive heat or moisture.
Do not open – there are no user serviceable parts inside.
Do not block the chassis vent slots or the fan inlet.
Do not operate in an explosive environment.

Introduction

This document describes the operation of the filament module for the MIPS system. The filament module drives one or two filaments with a number of programmable features described in this manual. This module is designed to drive low-voltage, high-current filaments. The general features are listed below:

- Two independent channels per module
- Filament supply voltage, 0.7 V to 5.0 V
- Filament current, programmable 0 to 12 A
- Filament power, 0 to 20 W
- Maximum power monitoring and control
- Programmable current ramp rate (0 to 1 A/s)
- External float voltage input up to 500 V (both channels float together)
- Two modes of operation: Icrtl (constant current) and IVcrtl (constant current with automatic supply voltage control to minimize power dissipation)
- Current cycling between two user-defined setpoints with programmable cycle count
- Emission current monitoring via DC bias channel 8 (resolution 2 μA)
- Programmable periodic parameter reporting to host computer
- USB host computer interface allowing control of all major parameters
- Real-time display of filament current, voltage, and power

The MIPS system supports two filament modules, allowing up to four filaments to be controlled by a single MIPS system.

Please refer to the MIPS operations manual for details on the general operation of MIPS and use of the user interface.

Description

The filament controller uses a point-of-load (POL) switching power supply to generate the filament supply voltage, which can be set between 0.7 V and 5.0 V. This supply voltage is applied to one side of the filament. The other (load) side of the filament is connected to a current source implemented using a power MOSFET. Filament current is monitored with a Hall-effect sensor. A feedback loop on the filament control module compares the measured current with the desired setpoint and adjusts the MOSFET gate drive to achieve the target value.

All voltages and currents are monitored using I2C Analog Devices ADCs and an Arduino DUE microcontroller. Output voltage and current setpoints are set using I2C Analog Devices DACs.

Galvanic isolation is provided between the microcontroller and the filament control electronics, allowing the user to supply a bias voltage and "float" the filament at up to 500 V. Both filament channels on a single module float at the same voltage.

The filament current can be cycled between two user-defined current setpoints (point 1 and point 2) for a programmable number of cycles. One cycle is defined as the current changing from point 1 to point 2 and back to point 1. When all cycles are complete the current returns to its original setpoint value. Cycling parameters are controlled from the Cycle menu or via host computer commands. The ramp rate parameter governs how quickly the current changes as it cycles between setpoints.

Emission current from the filament is monitored using DC bias output channel 8, which provides the bias voltage through a 10 kΩ current sense resistor. The resulting emission current is displayed in micro-amps with a resolution of 2 µA.

Filament Control Parameters

Selecting the Filament module from the MIPS main menu opens the Filament control parameters dialog, which displays filament channel status and allows control of key parameters. Each parameter is described below.

Page 1 Parameters

Channel

Selects the filament channel whose parameters are to be controlled and monitored.

Enable

Turns a filament channel on or off. Setting a channel from OFF to ON causes the filament current to ramp up at the user-defined ramp rate. Setting a channel from ON to OFF immediately sets the filament current to zero.

Current

The filament current setpoint in amps. The control electronics will ramp the filament current to this value at the rate defined by the Ramp rate parameter. The filament is held at this current level unless an over-power condition is detected.

Voltage

The filament supply voltage. The voltage required to generate the desired current depends on the filament's resistance, which varies with power dissipation. This parameter defines the voltage applied to one side of the filament and effectively sets the maximum achievable current. In Icrtl mode this parameter is set by the user; in IVcrtl mode it is controlled automatically by the firmware to minimize power dissipation in the current source. This voltage should be set high enough to achieve the required filament current but not so high as to dissipate excessive power in the current source.

Bias I (µA)

Display-only. Reports the emission current from the filament in micro-amps, calculated from the DC bias output channel 8 voltage across a 10 kΩ current sense resistor. Resolution is 2 µA.

Supply V *Filament V* *Filament I* *Power*

These four parameters are read-only monitored values. Supply V is the filament current source supply voltage applied to one side of the filament. Filament V is the voltage measured on the current source (load) side of the filament; the actual voltage across the filament is Supply V minus Filament V. Filament I is the measured filament current, read by the Hall-effect current sensor. Power is the calculated power dissipated in the filament: (Supply V − Filament V) × Filament I.

Next page

Displays the second page of the filament module dialog.

Return to main menu

Returns to the MIPS main menu.

Page 2 Parameters

Selecting Next page from the first page displays the following additional parameters.

Mode

Selects the operating mode: **Icrtl** or **IVcrtl**. In Icrtl mode the current is set by the current setpoint and the user manually defines the filament supply voltage. In IVcrtl mode the current is set by the current setpoint and the supply voltage is automatically adjusted by the firmware to minimize power dissipation in the current source.

Max power

Defines the maximum allowable power dissipation for the filament in watts. If this limit is exceeded the filament current is automatically reduced until the power falls below the limit.

Ramp rate

The rate of change of the filament current in amps per second (range 0 to 1 A/s). When the current setpoint is changed — either through the MIPS user interface or via a host computer command — this parameter limits the actual rate of current change, allowing the filament to be heated gradually. This parameter also governs the transition rate during current cycling.

Cal Supply V *Cal Filament V* *Cal Filament I*

These three selections calibrate the displayed voltage and current values and are used for system setup only. It should not be necessary to adjust these settings unless hardware components have been replaced. The calibration procedure requires a filament to be connected and the controller to be enabled; measured voltages and currents at the filament are needed to complete calibration.

Cycle menu
Opens the Cycle menu for controlling and monitoring current cycling parameters. When current cycling is active, do not adjust the filament current manually as this will produce unexpected results.

Save settings
Saves the complete filament module configuration to non-volatile memory on the filament module board. A status popup confirms the result. Parameter changes are not retained across power cycles or reboots unless this option is used.

Restore settings
Reloads the filament configuration from non-volatile memory. This is called automatically when the MIPS system powers up.

First page
Returns to the first page of the filament module dialog.

Return to main menu
Returns to the MIPS main menu.

Cycle Menu Parameters

Selecting Cycle menu from the second page displays the following cycling control parameters.

Enable cycling
Set to Yes to enable current cycling mode, or No to disable it. Ensure all cycling parameters are configured before enabling. When the programmed number of cycles completes, this parameter automatically resets to No.

Current 1
The first current setpoint in amps for the cycling mode (range 0.0–12.0 A).

Current 2
The second current setpoint in amps for the cycling mode (range 0.0–12.0 A).

Num cycles
The total number of current cycles to perform before cycling stops. One cycle is defined as the current changing from point 1 to point 2 and back to point 1 (range 1–1000).

Current cycle
Display-only. Shows the number of completed cycles when cycling is in progress.

Return filament menu
Returns to the filament parameters dialog.

Host Computer Interface Commands

All commands are ASCII text. Commands sent to the MIPS system are terminated with a carriage return (CR) or line feed (LF) character. All responses from the MIPS system are terminated with CR and LF.

After receiving a command MIPS responds with an ASCII ACK (0x06) followed by CR if the command was accepted, or an ASCII NAK (0x15) followed by "?" and CR if the command or its arguments were not understood. The "?" character is included to assist users working through a terminal emulator.

The *Chan* parameter used by all filament commands is the filament channel number. With one module installed the valid range is 1–2; with two modules installed the valid range is 1–4. All commands are case-sensitive.

General Commands

GFLENA,*Chan*

Returns the enabled status (ON or OFF) of the selected filament channel.

SFLENA,*Chan*,*State*

Enables or disables the selected filament channel. *State* = ON or OFF.

GFLI,*Chan*

Returns the current setpoint of the selected filament channel in amps (range 0.0–12.0 A).

GFLAI,*Chan*

Returns the actual measured current of the selected filament channel in amps (range 0.0–12.0 A).

SFLI,*Chan*,*Current*

Sets the current setpoint of the selected filament channel. *Current* = desired value in amps (range 0.0–12.0 A).

GFLRT,*Chan*

Returns the current ramp rate for the selected filament channel in amps per second (range 0–1 A/s).

SFLRT,*Chan*,*Rate*

Sets the current ramp rate for the selected filament channel. *Rate* = current change rate in amps per second (range 0–1 A/s).

GFLSV,*Chan*

Returns the filament supply voltage setpoint for the selected channel. This is the value defined by the user or set automatically by the system, depending on the operating mode.

GFLASV,*Chan*
Returns the actual measured filament supply voltage for the selected channel (range 0.7–5.0 V).

SFLSV,*Chan*,*Voltage*
Sets the filament supply voltage for the selected channel. *Voltage* = desired value in volts (range 0.7–5.0 V). Note: in IVcrtl mode this value is determined automatically by the firmware and this command has no effect.

GFLV,*Chan*
Returns the measured voltage on the load side of the selected filament channel. The actual voltage across the filament is Supply V minus this value (range 0.7–5.0 V).

GFLPWR,*Chan*
Returns the calculated power dissipated in the selected filament channel in watts (range 0–20 W). Power is calculated from the filament current and the voltage drop across the filament.

RFLPARMS,*Chan*,*Period*
Enables periodic reporting of filament parameters for the selected channel. Reports are issued every *Period* seconds in the format:
FIL#,current,voltage,power
where # is the filament channel number and current, voltage, and power are comma-separated measured values. Setting *Period* to 0 stops the reporting.

GFLECUR
Returns the emission current in micro-amps.

SFLSRES,*Value*
Sets the value of the current sense resistor used to monitor emission current, in ohms. This is a configuration command used during initial system setup and should not normally be adjusted. *Value* = the sensor resistance in ohms.

Current Cycling Commands

GFLP1,*Chan*
Returns the current setpoint for point 1 of the cycling mode for the selected channel (range 0.0–12.0 A).

SFLP1,*Chan*,*Current*
Sets the current setpoint for point 1 of the cycling mode for the selected channel. *Current* = desired value in amps (range 0.0–12.0 A).

GFLP2,*Chan*
Returns the current setpoint for point 2 of the cycling mode for the selected channel (range 0.0–12.0 A).

SFLP2,*Chan*,*Current*
Sets the current setpoint for point 2 of the cycling mode for the selected channel. *Current* = desired value in amps (range 0.0–12.0 A).

GFLCY,*Chan*
Returns the number of cycles configured for the selected channel. One cycle = current point 1 → point 2 → point 1 (range 1–1000).

SFLCY,*Chan*,*Count*
Sets the number of cycles to perform for the selected channel. *Count* = number of requested cycles (range 1–1000).

GFLENAR,*Chan*
Returns the cycling mode status for the selected channel. Returns NO if cycling is disabled or stopped. If cycling is active, returns the number of remaining cycles.

SFLENAR,*Chan*,*Status*
Enables or stops the cycling mode for the selected channel. *Status* = YES to enable and start cycling, NO to disable and stop cycling.

Table 1, Filament Host Interface Commands

Command	Parameters	Description
GFLENA	<Chan>	Returns the enabled status (ON or OFF) of the selected channel.
SFLENA	<Chan>,<State>	Enables or disables the selected channel. State = ON or OFF.
GFLI	<Chan>	Returns the current setpoint (0.0–12.0 A).
GFLAI	<Chan>	Returns the actual measured current (0.0–12.0 A).
SFLI	<Chan>,<Current>	Sets the current setpoint. Current = desired value in amps (0.0–12.0 A).
GFLRT	<Chan>	Returns the current ramp rate in A/s (0–1 A/s).
SFLRT	<Chan>,<Rate>	Sets the current ramp rate. Rate = change rate in A/s (0–1 A/s).
GFLSV	<Chan>	Returns the supply voltage setpoint (user-defined or system-controlled depending on mode).
GFLASV	<Chan>	Returns the actual measured supply voltage (0.7–5.0 V).
SFLSV	<Chan>,<Voltage>	Sets the supply voltage (0.7–5.0 V). No effect in IVcrtl mode.
GFLV	<Chan>	Returns the measured load-side voltage (0.7–5.0 V). Filament voltage = Supply V − this value.
GFLPWR	<Chan>	Returns the calculated power dissipated in the selected channel (0–20 W).
RFLPARMS	<Chan>,<Period>	Enables periodic reporting of FIL#,current,voltage,power every Period seconds. Period = 0 stops reporting.
GFLECUR		Returns the emission current in micro-amps (µA).
SFLSRES	<Value>	Sets the emission current sense resistor value in ohms. Factory configuration command; do not adjust unless hardware has been replaced.
GFLP1	<Chan>	Returns the cycling mode point 1 current setpoint (0.0–12.0 A).

SFLP1	<Chan>,<Current>	Sets the cycling mode point 1 current. Current = desired value in amps (0.0–12.0 A).
GFLP2	<Chan>	Returns the cycling mode point 2 current setpoint (0.0–12.0 A).
SFLP2	<Chan>,<Current>	Sets the cycling mode point 2 current. Current = desired value in amps (0.0–12.0 A).
GFLCY	<Chan>	Returns the configured number of cycles (1–1000).
SFLCY	<Chan>,<Count>	Sets the number of cycles to perform. Count = number of cycles (1–1000).
GFLENAR	<Chan>	Returns cycling mode status. NO = disabled/stopped; number = remaining cycles if active.
SFLENAR	<Chan>,<Status>	Enables or stops cycling. Status = YES to start, NO to stop.